Topic 1 [STL: stack class]

Problem statement: Write a program to create the following menu and manipulate Stack using **stack** class. Use the following methods to write the program. Also display appropriate message when required.

i) **push()** -method to push data
ii) **pop()** -method to pop data
iii) **top()** -method to display top element
iv) **empty()** -method to check whether stack is empty or not

**** Stack Menu ****
1. Push
2. Pop
3. Display
4. Exit
Enter your option:

Code	Output

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define newl cout<<endl
#define clr system("cls");
class Stack
{
   private:
   stack<int>ax;
   public:
   Stack()
   {

   }
   void Push(int x)
   {
      ax.push(x);
   }
   void Pop()
   {
      ax.pop();
   }
   void Display()
   {
      if (ax.size()==0)
      {
         cout<<"NO ELEMENT";newl;
         return;
      }
      cout<<ax.top();newl;
   }

};
void menu()
{
   cout<<"**** Stack Menu ****\n";
   cout<<"   "<<"1. Push\n";
   cout<<"   "<<"2. Pop\n";
   cout<<"   "<<"3. Display\n";
   cout<<"   "<<"4. Exit\n";
   cout<<"      "<<"Enter your option: ";
}
int main()
{
   Stack s;
   menu();
   while(1)
```

#Main Menu:

```
**** Stack Menu ****
  1. Push
  2. Pop
  3. Display
  4. Exit
    Enter your option:
```

#Push operation:

```
**** Stack Menu ****
  1. Push
  2. Pop
  3. Display
  4. Exit
    Enter your option: 1
Enter Value: 12
```

#Pop operation:

```
**** Stack Menu ****
  1. Push
  2. Pop
  3. Display
  4. Exit
    Enter your option: 2
Value removed successfully
Press Enter
```

#Display operation:

```
**** Stack Menu ****
  1. Push
  2. Pop
  3. Display
  4. Exit
    Enter your option: 3
20
```

```
  {
    int a;
    cin>>a;
    if(a==1)
    {
      cout<<"Enter Value: ";
      int x;cin>>x;
      s.Push(x);
      clr
      menu();
    }
    else if(a==2)
    {
      s.Pop();
      cout<< "Value removed successfully\nPress Enter";
      getchar();getchar();
      clr
      menu();
    }
    else if(a==3)
    {
      s.Display();
      getchar();getchar();
      clr
      menu();
    }
    else
    {
      break;
    }
  }

}
```

Topic 2 [STL: queue class]

Problem statement: Write a program to create the following menu to manipulate Queue using queue class. Use the following methods to write the program. Also display appropriate message when required.

i) **push()** -method to push data
ii) **pop()** -method to pop data
iii) **front()** -method to display front element
iv) **back()** -method to display rear element
v) **empty()** -method to check whether queue is empty or not

```
**** Queue Menu ****
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your option:
```

Code	Output

#include<bits/stdc++.h>
using namespace std;
#define _easy_
ios_base::sync_with_stdio(0);cin.tie(0);cout.tie(0);
#define ll long long int
#define str string
#define newl cout<<endl
#define clr system("cls");
class Queue
{
  private:
  queue<int>ax;
	#Main Menu:
	**** Queue Menu ****
	1. Enqueue
	2. Dequeue
	3. Display
	4. Exit
	Enter your option:

```
  public:
  Queue()
  {

  }
  void Enqueue(int x)
  {
    ax.push(x);
  }
  void Dequeue()
  {
    ax.pop();
  }
  void Display()
  {
    if (ax.size()==0)
    {
      cout<<"NO ELEMENT";newl;
      return;
    }
    cout<<"Front Element: "<<ax.front();newl;
    cout<<"Rear Element: "<<ax.back();newl;
  }

};
void menu()
{
  cout<<"**** Queue Menu ****\n";
  cout<<"   "<<"1. Enqueue\n";
  cout<<"   "<<"2. Dequeue\n";
  cout<<"   "<<"3. Display\n";
  cout<<"   "<<"4. Exit\n";
  cout<<"     "<<"Enter your option: ";
}
int main()
{
  Queue q;
  menu();
  while(1)
  {
    int a;
    cin>>a;
    if(a==1)
    {
      cout<<"Enter Value: ";
      int x;cin>>x;
      q.Enqueue(x);
      clr
      menu();
    }
    else if(a==2)
    {
      q.Dequeue();
      cout<<"Value removed successfully\nPress Enter";
      getchar();getchar();
      clr
      menu();
    }
    else if(a==3)
    {
      q.Display();
      getchar();getchar();
      clr
      menu();
    }
    else
    {
      break;
    }
```

#Enqueue operation:

```
**** Queue Menu ****
  1. Enqueue
  2. Dequeue
  3. Display
  4. Exit
    Enter your option: 1
Enter Value: 15
```

#Dequeue operation:

```
**** Queue Menu ****
  1. Enqueue
  2. Dequeue
  3. Display
  4. Exit
    Enter your option: 2
Value removed successfully
Press Enter
```

#Display operation:

```
**** Queue Menu ****
  1. Enqueue
  2. Dequeue
  3. Display
  4. Exit
    Enter your option: 3
Front Element: 25
Rear Element: 45
```

```
    }
}
```

Topic 3 [STL: deque class]

Problem statement: Extend the above menu with more features using **deque** class. Use the methods to given in the lecture note for **deque** class.

Code

```
#include<bits/stdc++.h>
using namespace std;
#define _easy_
ios_base::sync_with_stdio(0);cin.tie(0);cout.tie(0);
#define ll long long int
#define str string
#define P_B push_back
#define p_b pop_back
#define P_F push_front
#define p_f pop_front
#define newl cout<<endl
#define clr system("cls");
class Deque
{
   private:
   deque<int>ax;
   public:
   Deque()
   {

   }
   void Push_Back(int x)
   {
      ax.P_B(x);
   }
   void Push_Front(int x)
   {
      ax.P_F(x);
   }
   void Pop_Back()
   {
      ax.p_b();
   }
   void Pop_Front()
   {
      ax.p_f();
   }
   void Display()
   {
      if (ax.size()==0)
      {
         cout<<"NO ELEMENT";newl;
         return;
      }
      cout<<"Front Element: "<<ax.front();newl;
      cout<<"Rear Element: "<<ax.back();newl;
   }

};
void menu()
{
   cout<<"**** Deque Menu ****\n";
   cout<<"   "<<"1. Enqueue (Front)\n";
   cout<<"   "<<"2. Enqueue (Back)\n";
   cout<<"   "<<"3. Dequeue (Front)\n";
   cout<<"   "<<"4. Dequeue (Back)\n";
   cout<<"   "<<"5. Display\n";
```

Output

#Main Menu:

```
**** Deque Menu ****
  1. Enqueue (Front)
  2. Enqueue (Back)
  3. Dequeue (Front)
  4. Dequeue (Back)
  5. Display
  6. Exit
    Enter your option:
```

#Enqueue Front operation:

```
**** Deque Menu ****
  1. Enqueue (Front)
  2. Enqueue (Back)
  3. Dequeue (Front)
  4. Dequeue (Back)
  5. Display
  6. Exit
    Enter your option: 1
Enter Value: 15
```

#Enqueue Back operation:

```
**** Deque Menu ****
  1. Enqueue (Front)
  2. Enqueue (Back)
  3. Dequeue (Front)
  4. Dequeue (Back)
  5. Display
  6. Exit
    Enter your option: 2
Enter Value: 20
```

#Dequeue Front operation:

```
**** Deque Menu ****
  1. Enqueue (Front)
  2. Enqueue (Back)
  3. Dequeue (Front)
  4. Dequeue (Back)
  5. Display
  6. Exit
    Enter your option: 3
Value removed successfully from front
Press Enter
```

```
   cout<<"   "<<"6. Exit\n";
   cout<<"      "<<"Enter your option: ";
}
int main()
{
   Deque dq;
   menu();
   while(1)
   {
      int a;
      cin>>a;
      if(a==1)
      {
         cout<<"Enter Value: ";
         int x;cin>>x;
         dq.Push_Front(x);
         clr
         menu();
      }
      else if(a==2)
      {
         cout<<"Enter Value: ";
         int x;cin>>x;
         dq.Push_Back(x);clr
         menu();
      }
      else if(a==3)
      {
         dq.Pop_Front();
         cout<<"Value removed successfully from front\nPress
Enter";
         getchar();getchar();
         clr
         menu();
      }
      else if(a==4)
      {
         dq.Pop_Back();
         cout<<"Value removed successfully from back\nPress
Enter";
         getchar();getchar();
         clr
         menu();
      }
      else if(a==5)
      {
         dq.Display();
         getchar();getchar();
         clr
         menu();
      }
      else
      {
         break;
      }
   }
}
```

#Dequeue Back operation:

```
**** Deque Menu ****
  1. Enqueue (Front)
  2. Enqueue (Back)
  3. Dequeue (Front)
  4. Dequeue (Back)
  5. Display
  6. Exit
    Enter your option: 4
Value removed successfully from back
Press Enter
```

#Display operation:

```
**** Deque Menu ****
  1. Enqueue (Front)
  2. Enqueue (Back)
  3. Dequeue (Front)
  4. Dequeue (Back)
  5. Display
  6. Exit
    Enter your option: 5
Front Element: 26
Rear Element: 55
```

Topic 4 [STL: list class]

Problem statement: Write a program to create and manipulate linked list using **list** class and its following methods to

i) insert 8 integers using **push_back()** method
ii) insert two elements using **push_front()** method
iii) display all the elements of the list in forward direction with user-defined Display() method using **begin()** and **end()** methods and iterator
iv) display all the elements of the list in reverse direction with a user-defined DisplayRev() method using **rbegin()** and rend() method and iterator
v) display front element using **front()** method
vi) display back element using **back()** method
vii) delete front element using **pop_back()** method
viii) delete front element using **pop_front()** method
ix) search an element **x** using **find()** method
x) insert a new element **x** before an existing element **y** using **insert()** method
xi) insert a new element **x** after an existing element **y** using **insert()** method
xii) count a particular element **x**
xiii) count a elements with condition using predicate function
xiv) delete a particular element **x** with **erase()** method
xv) delete first 4 elements with **erase()** method
xvi) delete a particular element **x** with **remove()** method
xvii) delete a elements with condition using **remove_if()** method using predicate function
xviii) assign elements from another list using **assign()** method
xix) assign elements from an array using **assign()** method
xx) sort the list using **sort()** method
xxi) Delete consecutive similar elements using **unique()** method

Code	Output

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define ev(x) x%2==0
#define od(x) x%2==1
#define mod0(x,y) x%y==0
#define P_B push_back
#define p_b pop_back
#define P_F push_front
#define p_f pop_front
#define yes_or_no cout<<"YES ":cout<<"NO "
#define newl cout<<endl
int Even_count(list<int>&li)
{
   int x=0;
   for(auto i:li)
   {
      if(ev(i))
      {x++;}
   }
   return x;
}
bool is_odd(int x)
{ return od(x); }
class List
{
   private:
   list<int>ax;
   public:
   List()
   {   }
   void Push_Back(int x)
   { ax.P_B(x); }
   void Push_Front(int x)
   { ax.P_F(x); }
   void Pop_Back()
```

```
{ ax.p_b(); }
void Pop_Front()
{ ax.p_f(); }
void Display()
{
   if(ax.size()==0)
   { cout<<"NO ELEMENT\n";
     return; }
   auto i=ax.begin(); auto j=ax.end();
   while(i!=j)
   { cout<<*i<<" "; i++; }
   newl; }
void DisplayRev()
{
   if(ax.size()==0)
   { cout<<"NO ELEMENT\n";
     return; }
   auto i=ax.rbegin(),j=ax.rend();
   while(i!=j)
   { cout<<*i<<" "; i++; }
   newl; }
int Front_show()
{
   if (ax.size()==0)
   { cout<<"NO ELEMENT";newl;
     return 0; }
   return ax.front();
}
void Back_show()
{
   if (ax.size()==0)
   { cout<<"NO ELEMENT";newl;
     return ; }
   cout<<"Rear Element: "<<ax.back();newl;
}
int Find(int x)
{ int y=(ax.begin(),ax.end(),x);
  return y; }
void Insert(int i,int x)
{ auto it=ax.begin();advance(it,i);
   ax.insert(it,x);Display(); }
void Count(int x)
{
   if(Find(x)==0)
   { cout<<x<<"is not found\n";return; }
   cout<<count(ax.begin(),ax.end(),x);newl;
}
void Even()
{ cout<<Even_count(ax);newl; }
void Delete(int i)
{ auto z=ax.begin();advance(z,i);
  ax.erase(z);Display(); }
void Sort()
{ ax.sort(); }
void Unique()
{ auto it=unique(ax.begin(),ax.end());
  ax.erase(it,ax.end());
  Display(); }
void Remove(int x)
{ auto b=remove(ax.begin(),ax.end(),x);
  ax.erase(b,ax.end());
  Display(); }
void Remove_If()
{ auto a=remove_if(ax.begin(),ax.end(),is_odd);
  ax.erase(a,ax.end());
  Display(); }
void Assign(list<int>B)
{ ax.assign(B.begin(),B.end());
  Display(); }
```

```
   void Array_assign(int A[],int n)
   { ax.assign(A,A+n); }
};
int main()
{
   int A[]={10,64,25,36,78,64,69,70,64,67,67,62,62};
   int n=13;
   List li;
   for(int i=1;i<=8;i++)
   {
      li.Push_Back(A[i-1]);
   }
   li.Display();  // i)
   li.Push_Front(A[8]);li.Push_Front(A[9]);
   li.Display();  // ii)
   li.Display();   // iii)
   li.DisplayRev(); // iv)
   cout<<"Front Element: ";
   cout<<li.Front_show();newl;   // v)
   li.Back_show();  // vi)
   li.Pop_Front();li.Display();    // vii)
   li.Pop_Back();li.Display();   // viii)
   li.Find(64)?yes_or_no;li.Find(50)?yes_or_no; newl;   // ix)
   li.Insert(2,75);   // x)
   li.Insert(4,110);    // xi)
   li.Count(64);     // xii)
   li.Even();   // xiii)
   li.Delete(36);   // xiv)
   for(int i=1;i<=4;i++)
   {
      li.Delete(li.Front_show());
   }    // xv)
   li.Remove(78);   // xvi)
   li.Remove_If();   // xvii)
   list<int>B={15,25,51,45,60,80,64};
   li.Assign(B); // xviii)
   li.Array_assign(A,n);   // xix)
   li.Sort();li.Display();   // xx)
   li.Unique();   // xxi)

}
```

i)
10 64 25 36 78 64 69 70
ii)
67 64 10 64 25 36 78 64 69 70
iii)
67 64 10 64 25 36 78 64 69 70
iv)
70 69 64 78 36 25 64 10 64 67
v)
Front Element: 67
vi)
Rear Element: 70
vii)
64 10 64 25 36 78 64 69 70
viii)
64 10 64 25 36 78 64 69
ix)
YES NO
x)
64 10 75 64 25 36 78 64 69
xi)
64 10 75 64 110 25 36 78 64 69
xii)
3
xiii)
7
xiv)
64 10 75 64 110 25 78 64 69
xv)
10 75 64 110 25 78 64 69
75 64 110 25 78 64 69
64 110 25 78 64 69
110 25 78 64 69
xvi)
110 25 64 69
xvii)
110 64
xviii)
15 25 51 45 60 80 64
xix)
10 64 25 36 78 64 69 70 64 67 67 62 62
xx)
10 25 36 62 62 64 64 64 67 67 69 70 78
xxi)
10 25 36 62 64 67 69 70 78